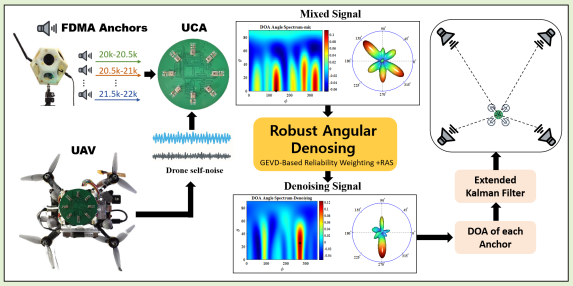
LUNA: A Low-SNR Acoustic Localization and Orientation Framework for UAVs

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***Abstract*— Accurate indoor localization and stable attitude estimation of unmanned aerial vehicles (UAVs) are essential for autonomous navigation, obstacle avoidance, and mission execution. However, existing indoor localization solutions, such as Light Detection and Ranging (LiDAR), vision-based systems, and Ultra-Wideband (UWB) technologies, may fail in certain indoor scenarios. In this context, as one implementation within the indoor localization framework, acoustic localization has attracted increasing attention due to its low cost, high accuracy, and insensitivity to ambient lighting conditions. This paper presents an acoustic localization system named LUNA, Low-SNR UAV Localization via Acoustics, which achieves accurate indoor position and orientation estimation using commercial loudspeakers and an eight-element circular microphone array. Loudspeakers deployed in the environment emit near-ultrasonic signals, which are received by the UAV-mounted microphone array. Under strong UAV self-noise conditions, a reduced angular-spectrum method is applied to perform direction-of-arrival (DoA) estimation, and position and orientation are jointly optimized by minimizing an angle-of-arrival (AoA) related objective function. Experiments conducted in real indoor environments demonstrate that LUNA achieves a mean position error of 0.089 m, representing an improvement of approximately 30% over commercial UWB systems and 80% over traditional time-difference-of-arrival (TDoA) methods. The mean orientation error is 0.56°, demonstrating the system's high precision, robustness, and strong potential for real-world indoor UAV navigation applications.**



***Index Terms*— Acoustic indoor localization, drone, low-SNR, direction-of-arrival (DoA), angle-of-arrival (AoA), microphone array.**

I. Introduction

In recent years, indoor localization and navigation of unmanned aerial vehicles (UAVs) have played an important role in enabling autonomous indoor operations such as warehouse inventory [1], security inspection [2], and emergency response [3]. However, in indoor environments, signals from Global Navigation Satellite Systems (GNSSs) [4], [5], [6], such as the Global Positioning System (GPS) [7] and BeiDou Navigation Satellite System (BDS) [8], are blocked by walls and roofs, making satellite-based positioning infeasible [9]. Therefore, achieving centimeter-level indoor localization has become a current research hotspot and an essential technology for ensuring the safe and autonomous operation of UAVs [10].

Current indoor localization techniques leverage a range of sensing modalities, including Wireless Fidelity Wi-Fi [11], [12], Bluetooth Low Energy (BLE) [13], [14], Ultra-Wideband (UWB) [15], [16], Radio-Frequency Identification (RFID) [17], visible light [18], and acoustics. While each has its merits in specific scenarios, notable limitations arise when applied to UAVs. Radio-frequency technologies, represented by Ultra-Wideband (UWB), are currently among the most widely adopted solutions for indoor UAV localization. However, in certain indoor environments, such as metal-rich or strongly multipath scenarios, their performance may degrade due to signal attenuation and excessive delay spread [19]. Vision [20] or LiDAR-based methods [21], [22] can deliver fine-grained pose estimation, yet their performance degrades significantly under poor illumination, occlusion, or smoke, and their computational cost is often prohibitive for lightweight UAV platforms [23], [24], [25]. In contrast, acoustic localization is low-cost, easy to deploy, and naturally suited to GPS-denied indoor environments, and has therefore received growing interest in recent years [26], [27], [28].

Acoustic localization methods are generally categorized into time of arrival (ToA), time difference of arrival (TDoA), and direction of arrival (DoA) [29], [30]. Most existing UAV acoustic systems adopt ultrasonic ToA/TDoA schemes using dedicated ultrasonic transducers that emit high-frequency pulses [31], [32]. Although such designs can achieve accurate ranging with high signal-to-noise ratio (SNR), they rely on high-voltage driver circuits and strict anchor–UAV clock synchronization, and the additional transducer modules increase

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payload, power consumption, and deployment complexity, especially when multiple sensors must be distributed around the airframe.

By comparison, DoA-based methods estimate the angle of the incoming wavefront using microphone arrays, can operate under asynchronous conditions, and are intrinsically more robust to Doppler shifts induced by UAV motion. Recent studies on ground robots and mobile devices have shown that localization can be achieved using commodity loudspeakers and wideband microphone arrays [33], [34], [35], enabling accurate indoor navigation with only a passive audio front-end on the robot side. Inspired by these works, we employ consumer-grade loudspeakers as anchors together with a lightweight onboard array of MEMS microphones to realize DoA-based acoustic localization on UAVs. This architecture eliminates the need for dedicated ultrasonic hardware, simplifies system integration, and keeps the UAV payload small. In addition, because the microphones retain sensitivity in the audible band, the same sensing platform can be reused for other audio perception tasks such as human-voice source localization, speech recognition, and speech-based human–UAV interaction [36], [37], [38]. However, realizing stable, real-time, and high-accuracy acoustic localization for UAVs in practical indoor environments still faces three key challenges:

Challenge 1: Strong self-noise from the UAV. Aerodynamic noise generated during flight together with harmonic energy around the blade-passing frequency (BPF) is extremely strong and partially overlaps the positioning band [39]. This onboard self-noise forms a high-energy, broadband interference field near the microphone array, markedly degrading the SNR. It is a distinctive bottleneck for UAV acoustic localization compared with other mobile platforms.

Challenge 2: Joint estimation of position and orientation. In complex mission scenarios, UAV autonomy relies on the simultaneous acquisition of precise spatial coordinates and stable attitude information, which jointly support accurate path planning, steady hovering, and effective obstacle avoidance.

Challenge 3: Multipath interference in indoor environments. Indoor spaces are rich in reflectors (walls, racks, pillars), which induce severe multipath and reverberation. Reflected components superimpose with the line-of-sight path and bias both ToA/TDoA and DoA estimates, degrading localization accuracy, especially in large or highly reflective spaces.

In this paper, we propose a cooperative indoor UAV localization scheme. Loudspeakers deployed in the environment transmit linear frequency-modulated (LFM) chirps, which are received and processed onboard by a microphone-array-equipped UAV. We perform self-noise suppression for low-SNR chirps via a weighted angular-spectrum subtraction method, enabling robust identification of the bearing to each anchor even under strong UAV self-noise. We then estimate the UAV's position and orientation using DoA-based bearing measurements, providing precise spatial coordinates together with stable attitude for complex missions.

Our contributions can be summarized as follows:

- We develop an active acoustic indoor UAV localization system in which four anchors transmit linear frequency-modulated (LFM) signals under a frequency-division multiple-access (FDMA) scheme, and an onboard microphone array receives them. A DoA-only strategy eliminates cross-device synchronization and dedicated ranging links.
- We propose a Reduced Angle Spectrum (RAS) method for DoA estimation that suppresses UAV self-noise by exploiting spatial-coherence disparities and sub-band weighting. The RAS algorithm performs angular-spectrum subtraction and adaptive reweighting to recover anchor bearing peaks under low SNR conditions, significantly improving DoA estimation robustness against BPF interference and its harmonics, while multipath reflections are simultaneously attenuated during the denoising process.
- We design a DoA-based joint position–and–orientation estimator built upon a circular-residual objective with robust penalties (Huber/hinge) and derivative-free optimization (Nelder–Mead with multi-start). Confidence values derived from the denoised spectrum are used to weight the observations, enabling stable and real-time pose estimation.
- We implemented the LUNA system, and experiments demonstrate that it can achieve high-accuracy UAV position and orientation estimation even under low SNR conditions. In indoor-scale environments, the average positioning error is 0.089 m, representing an improvement of about 30% compared with UWB systems and about 80% compared with traditional TDoA algorithms. The average orientation estimation error is 0.56°, indicating that the LUNA system provides reliable localization accuracy in indoor environments.

II. Related Works

A. UAV indoor acoustic localization

Current UAV indoor localization techniques based on acoustics can be broadly divided into two categories: **passive** and **active** methods.

Passive. Basiri et al. used teammate UAV engine noise as the source and a small triangular onboard array to estimate relative bearing via TDOA/coherence analysis, achieving an angle RMSE of $\sim 1.39^\circ$ in a $6\,\mathrm{m} \times 3.5\,\mathrm{m} \times 3\,\mathrm{m}$ indoor space. While no external emitters are needed, performance depends strongly on the spectral characteristics of UAV self-noise, overlap with ambient or other UAV noises and indoor multipath/reverberation can markedly degrade stability and accuracy [40]. Sun et al. proposed an acoustic–inertial localization approach for UAV tracking, in which ground-deployed microphones capture the UAV's self-generated noise as the acoustic signal for position estimation. Under line-of-sight (LOS) conditions, the method achieved a localization accuracy of 1.43 m [41].

Active. Active methods deploy controllable signals in the environment and generally maintain higher accuracy at low SNR. Representative ultrasonic TOA systems include APNS, PILOT, and RAIL [42]. PILOT targets high-precision 3-D UAV localization [43] using active ultrasound ranging and a three-stage mechanism that mitigates multipath and geometric

configuration errors, reporting a mean 3-D error $< 1.2\,\mathrm{cm}$ with a substantially reduced Z-axis error versus baseline. RAIL [32] introduces hybrid FH-CDMA (frequency hopping + code division) to combat indoor fading and signal separation issues, achieving a mean 3-D error of $\sim 1.5\,\mathrm{cm}$ in a $5\,\mathrm{m} \times 5\,\mathrm{m} \times 4\,\mathrm{m}$ corridor–over $2\times$ better than baseline FH-CDMA. Demircali et al. proposed an indoor warehouse system that integrates UAV autonomous flight with acoustic positioning, which the positioning error is 4–$5\,\mathrm{cm}$ [44].

As summarized in Table I, existing acoustic localization systems exhibit a clear trade-off between accuracy, synchronization requirements, and system complexity. Passive methods, such as AIM [41], avoid external emitters and achieve moderate accuracy but are highly sensitive to UAV self-noise and indoor reflections. In comparison, active acoustic localization has become the mainstream approach for UAV indoor positioning, as it enables centimeter-level precision by transmitting designed ultrasonic signals through synchronized anchors, as seen in systems such as APNS, PILOT, and RAIL. However, despite their outstanding accuracy, these active systems typically rely on specialized hardware, tight synchronization, and complex deployment procedures, which significantly increase cost and limit their scalability to lightweight UAV platforms. Consequently, there is a growing need for synchronization-free, low-complexity acoustic localization methods that can maintain robustness under strong UAV self-noise and low-SNR indoor conditions.

B. UAV sound source localization under low SNR

Using commercial loudspeakers and microphones instead of dedicated ultrasonic transducers can greatly reduce cost and deployment complexity while maintaining the benefits of active acoustic probing. However, UAV self-noise and indoor multipath interference severely degrade the SNR, making robust DoA estimation under low SNR a critical challenge for practical UAV acoustic localization systems.

To address extremely low SNR dominated by rotor noise, several array-processing methods improve DOA robustness. Salvati et al. combined diagonal-unloading (DU) beamforming with L1-norm frequency fusion (NORT) [45], maintaining $< 5^\circ$ DOA error at SNR as low as $-19\,\mathrm{dB}$, albeit with a relatively complex 8-mic uniform circular array (UCA) suspended beneath the UAV–less convenient indoors. Lin Wang & Cavallaro compared beamforming, blind source separation (BSS), and time-frequency (TF) filtering [46], showing that TF and BSS can outperform baselines at $-15\,\mathrm{dB}$ to $-30\,\mathrm{dB}$ SNR; TF often requires a direction prior to build filters, while BSS assumes quasi-static mixing, limiting applicability during dynamic flight. Furukawa et al. proposed GEVD-MUSIC with adaptive estimation of the noise correlation matrix (NCM) for spatial whitening [47]; it substantially improves the F-value at SNR $\geq -5\,\mathrm{dB}$, but accuracy degrades at SNR $\leq -10\,\mathrm{dB}$.

To overcome these limitations, we propose a DoA-based active acoustic localization framework tailored for indoor UAVs operating under low-SNR conditions. To ensure reliable acoustic probing in noisy environments, a linear frequency-modulated (LFM) chirp is adopted as the transmitted waveform. The LFM signal provides high time–frequency resolution and strong correlation properties, making it more resilient to multipath interference and noise contamination than narrowband tones.

As shown in Fig. 1, we propose the LUNA framework with a RAS front end: RAS exploits spatial-coherence disparities and sub-band weights to subtract and reweight angular spectra, enabling recovery of anchor bearing peaks under low SNR despite harmonics. For the back end, we develop a DoA-only joint position-heading estimator using circular angular residuals, robust penalties, and derivative-free optimization, delivering real-time and stable heading and position estimates. Compared with synchronized ultrasonic TOA systems, our approach lowers hardware complexity and power; compared with passive methods, it attains higher robustness under strong self-noise and multipath.

III. System Design

The proposed localization system, referred to as **LUNA**, consists of two main components. On the one hand, a lightweight uniform circular microphone array is mounted on the UAV, acting as the receiving terminal. On the other hand, four spatially distributed anchor nodes are deployed in the indoor environment to serve as active acoustic transmitters. Departing from conventional passive frameworks, our system allows the UAV to collect and store the acoustic transmissions from the anchor nodes during flight. This design requires no additional synchronization or communication between anchors and the UAV, thus achieving a low-cost, low-power, and fully autonomous localization scheme. At the current stage, offline processing is adopted primarily to ensure flexibility for algorithm validation and robustness testing under various flight conditions. The recorded dataset also enables detailed performance analysis and parameter tuning. However, the same framework can be seamlessly extended to real-time onboard computation in future implementations, as the DoA-based estimation only depends on locally captured acoustic signals and does not require inter-device synchronization.

The onboard microphone array adopts a planar geometry, where all microphones lie on the same array plane. In practical deployments, the UAV and the anchor nodes are not required to lie on the same horizontal plane, and height differences may exist between them. However, for a planar array, the array response mainly depends on the projection of the incoming wave direction onto the array plane, making azimuth estimation generally more reliable than elevation estimation.

Therefore, in this work we focus on azimuth estimation and use the azimuth measurements from multiple anchors to perform 2D horizontal localization, the positioning error can be constrained within the centimeter level under typical indoor conditions. This configuration effectively captures horizontal position and orientation while maintaining the system's lightweight and low-power advantages.

Considering the strong interference from UAV rotor self-noise and multipath reflections commonly encountered in indoor environments, we adopt the LFM signal as the probing waveform. Chirp signals exhibit excellent autocorrelation

TABLE I: Comparison of representative UAV indoor acoustic localization systems.

System	Method	Architecture	Accuracy	Sync.	Complexity	Cost
APNS [42]	TOA + TDMA	UAV receives	<0.1 m	Yes	High	High
PILOT [43]	FHSS TOA	UAV receives	≈0.012 m	Yes	High	High
RAIL [32]	FH-CDMA TOA + Z-Sensor	UAV receives	≈0.015 m	Yes	High	High
AIM [41]	Acoustic-Inertial	UAV transmits	1.43 m	Yes	Low	Medium
LUNA (Ours)	**RAS + AOA**	**UAV receives**	**0.089 m / 0.56°**	**No**	**Low**	**Low**

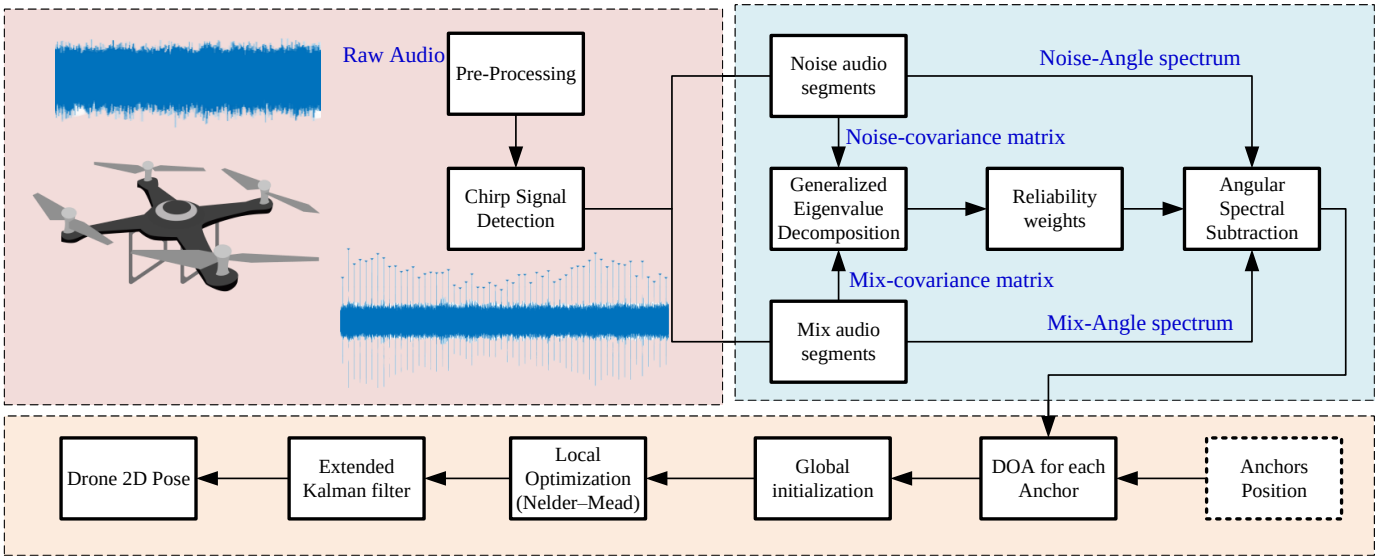


Fig. 1: **System architecture of the proposed LUNA acoustic localization framework.** The process begins with raw audio acquisition from the UAV, followed by pre-processing and chirp-signal detection. Noise and mixed audio segments are used to compute covariance matrices, which are processed through generalized eigenvalue decomposition (GEVD) and angular spectral subtraction to obtain reliable DoA estimates. By combining the DoA information from each anchor and their known coordinates, the UAV's 2D position and orientation are estimated. Finally, an extended Kalman filter (EKF) fuses the sequential localization results with system dynamics to achieve smooth attitude estimation.

properties and robustness against noise, which ensures reliable estimation performance even under low-SNR conditions. Furthermore, based on the analysis of UAV noise spectral characteristics, we observe that although rotor noise persists in the near-ultrasonic band (17-23 kHz), its power markedly decreases in the 20-22 kHz sub-band, as shown in Fig. 2. Therefore, this sub-band is selected as the primary operational frequency range, balancing propagation characteristics, energy efficiency, and noise suppression.

Formally, the chirp signal transmitted by the i-th anchor node can be defined as

$$s_i(t) = \cos\Big(2\pi\big(f_0 t + \tfrac{B}{2T}t^2\big)\Big), \quad 0 \le t \le T \tag{1}$$

where f_0 is the starting frequency, B denotes the sweep bandwidth, and T is the chirp duration. The linear frequency sweep property of $s_i(t)$ facilitates subsequent matched filtering and enables high-resolution AOA estimation, which forms the foundation of the proposed acoustic localization framework.

To avoid signal interference among multiple anchor nodes operating concurrently, orthogonal resource allocation is required. Common multiple access methods include TDMA (time-division multiple access), CDMA (code-division multiple access), and FDMA (frequency-division multiple access). Considering the real-time requirements of UAV localization and the limited onboard computational resources, this work

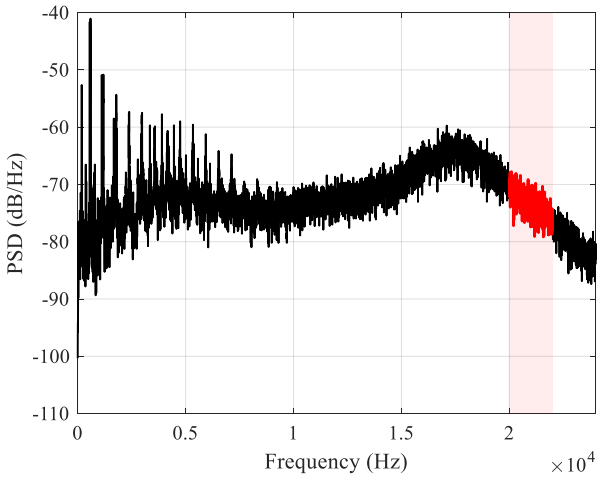


Fig. 2: **Mean PSD of UAV noise**. The 20-22 kHz sub-band is highlighted in red.

adopts the FDMA scheme, where each anchor is assigned a non-overlapping sub-band within the selected frequency range.

Specifically, in the chosen operational band of **20–22 kHz**, the frequency is uniformly divided into sub-bands of width $\Delta f = 0.5$ kHz. As an illustrative implementation, each sub-

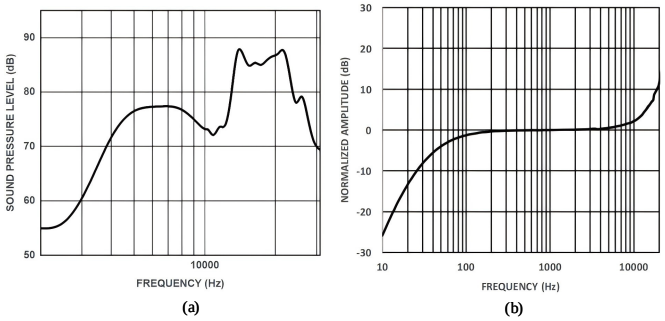


Fig. 3: **Frequency response of the acoustic hardware used in the system.** (a) Measured frequency response of the employed piezoelectric tweeter (1V, 1m). (b) Typical frequency response of the ICS-43434 MEMS microphone.

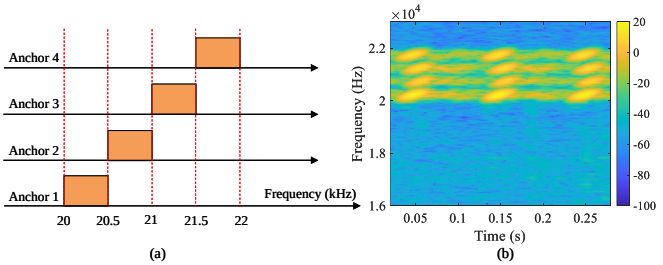


Fig. 4: **FDMA frequency allocation across anchors**. (a) Within the total available acoustic bandwidth of 20–22 kHz, the system assigns four anchors to four non-overlapping 0.5 kHz sub-bands, enabling parallel transmission without mutual interference. (b) Time-frequency spectrogram of the microphone's received signal, showing the received chirp packets from four anchors operating concurrently.

band is allocated to one anchor node. The starting frequency of the i-th anchor node can be expressed as

$$f_i = f_{\min} + (i-1)\Delta f \tag{2}$$

where $f_{\min} = 20$ kHz and $i = 1, \ldots, 4$ denotes the anchor index.

In our implementation, the acoustic transmitter is a commercial piezoelectric tweeter, while the receiver employs ICS-43434 digital MEMS microphones. Fig. 3 shows the frequency responses of the transmitter and the microphone. The measured response indicates that the tweeter provides usable acoustic output mainly in the range of approximately 10-30 kHz. The microphone array is sampled at 48 kHz. According to the Nyquist sampling theorem [48], the theoretical upper bound of the observable acoustic spectrum is therefore 24 kHz. Our experimental measurements further indicate that the employed microphones can still reliably capture acoustic signals in the 20-24 kHz range. In this work, a representative four-anchor configuration is adopted for experimental validation, corresponding to the sub-bands $[20.0, 20.5]$, $[20.5, 21.0]$, $[21.0, 21.5]$, and $[21.5, 22.0]$ kHz, respectively. This configuration is chosen to provide sufficient geometric diversity in a two-dimensional scenario and does not represent a fundamental limitation of the proposed framework. In principle, a larger number of anchors can be supported by further subdividing the available acoustic bandwidth or extending the operational frequency range, at the cost of a bandwidth–SNR trade-off, since narrower sub-bands reduce the effective signal energy per anchor, while extending to higher frequencies is constrained by the bandwidth characteristics of commercial speakers, microphone sensitivity, and sampling-rate limitations.

The specific FDMA allocation across anchors is illustrated in Fig. 4. This FDMA-based frequency allocation ensures clear signal separation among anchors and effectively eliminates co-channel interference, thereby providing reliable conditions for subsequent AOA estimation.

A. Microphone Array Signal Model

The UAV is equipped with a UCA of $M = 8$ microphones evenly placed on a circle of radius r. The location of the m-th microphone is defined in the UAV body coordinate system as

$$\mathbf{r}_m = [r_{m,x},\, r_{m,y},\, 0]^{\mathrm{T}}, \quad m = 1, 2, \ldots, M \tag{3}$$

which indicates that all microphones lie on the same planar array. The overall array geometry can therefore be written as

$$\mathbf{R} = [\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_M] \tag{4}$$

When the i-th anchor node located in the far-field emits an acoustic signal, the received microphone signal vector at time t is expressed as

$$\mathbf{x}(t) = [x_1(t), x_2(t), \cdots, x_M(t)]^{\mathrm{T}} \tag{5}$$

It consists of the desired acoustic signal vector and the noise vector, i.e.,

$$\mathbf{x}(t) = \mathbf{s}(t) + \mathbf{n}(t) \tag{6}$$

where $\mathbf{s}(t) = [s_1(t), s_2(t), \ldots, s_M(t)]^{\mathrm{T}}$ represents the desired acoustic signal from the i-th anchor, and $\mathbf{n}(t)$ denotes the additive noise including UAV self-noise and environmental noise.

After applying the short-time Fourier transform (STFT), the frequency-domain representation is given by

$$\mathbf{X}_i(n,k) = \mathbf{a}(\theta_i, \varphi_i, f_k) S_i(n,k) + \mathbf{N}(n,k) \tag{7}$$

where n and k denote the time-frame and frequency-bin indices of the STFT, respectively, and f_k is the physical frequency corresponding to the k-th bin. $\mathbf{a}(\theta_i, \varphi_i, f_k) \in \mathbb{C}^{M\times 1}$ is the steering vector of the array that characterizes the phase response corresponding to the azimuth θ_i and elevation φ_i. $S_i(n,k)$ is the spectrum of the i-th anchor signal, and $\mathbf{N}(n,k)$ is the noise vector across microphones. Although only azimuth information is used in the final localization stage, the received signal model is intentionally kept in the general three-dimensional DOA form. This formulation avoids implicitly enforcing $\varphi = 0$ and helps prevent height differences, slight UAV tilting, or multipath propagation effects from being incorrectly absorbed into the azimuth estimation.

Under the far-field approximation, assuming plane-wave incidence and neglecting body scattering, the m-th entry of

the steering vector for the signal from anchor i at frequency f_k can be written as

$$a_m(\theta_i, \varphi_i, f_k) = \exp(-j2\pi f_k \tau_{i,m}) \tag{8}$$

where the propagation delay to the m-th microphone is

$$\tau_{i,m} = \frac{1}{c}(\mathbf{r}_m - \mathbf{r}_{\text{ref}})^{\mathrm{T}}\mathbf{u}(\theta_i, \varphi_i) \tag{9}$$

and the unit direction vector $\mathbf{u}(\theta, \varphi)$ is defined as

$$\mathbf{u}(\theta, \varphi) = \begin{bmatrix} \cos\varphi\cos\theta \\ \cos\varphi\sin\theta \\ \sin\varphi \end{bmatrix} \tag{10}$$

Here, c is the speed of sound, $\mathbf{r}_m$ is the position of the m-th microphone, $\mathbf{r}_{\text{ref}}$ is the reference point (array center or a reference microphone), and f_k denotes the frequency of the k-th bin.

B. Low-SNR Angle-of-Arrival Estimation

From (5), the Short Time Fourier Transform (STFT) output of the q-th microphone at the n-th frame and the k-th frequency bin is denoted as

$$X_q(n,k) = |X_q(n,k)|e^{j\Phi_q(n,k)} \tag{11}$$

where $|X_q(n,k)|$ represents the amplitude spectrum and $\Phi_q(n,k)$ denotes the phase. For microphone channels q and q', the cross-power spectrum at (n,k) is defined as

$$C_{q,q'}(n,k) = X_q(n,k)X_{q'}^*(n,k) \tag{12}$$

By performing cross-power spectrum normalization between channels q and q', we obtain

$$\tilde{C}_{q,q'}(n,k) = \frac{X_q(n,k)X_{q'}^*(n,k)}{|X_q(n,k)X_{q'}^*(n,k)|} \tag{13}$$

which removes the amplitude component and preserves only the relative phase information. The instantaneous inter-channel phase difference can thus be written as

$$\Delta\Phi_{q,q'}(n,k) = \Phi_q(n,k) - \Phi_{q'}(n,k) \tag{14}$$

where $\Phi_q(n,k)$ and $\Phi_{q'}(n,k)$ denote the phases of the q-th and q'-th microphone signals, respectively.

This phase difference reflects the propagation path difference of the acoustic wave from the source to different microphones under the given array geometry, and it forms the basis for subsequent DOA estimation.

Although the normalized cross-power spectrum in (12)–(13) preserves the inter-channel phase difference, it suffers from two drawbacks under low-SNR conditions: (i) geometrical symmetry of the array may lead to front–back ambiguity since the amplitude cues are completely discarded, and (ii) spurious peaks may arise because frequency bins dominated by noise are given the same unit weight as reliable ones.

To address these issues, we propose an improved cross-power spectrum formulation that incorporates both a coherence-based weighting factor and a phase transform (PHAT)-λ truncation strategy, where PHAT denotes the phase-only weighting scheme commonly used in generalized cross-correlation (GCC) methods [49], defined as

$$\tilde{R}_{q,q'}(n,k) = w_{q,q'}(n,k)^{\alpha} \cdot \frac{X_q(n,k)X_{q'}^*(n,k)}{|X_q(n,k)X_{q'}^*(n,k)| + \lambda} \tag{15}$$

$$w_{q,q'}(n,k) = \frac{\left|\mathbb{E}\{X_q(n,k)X_{q'}^*(n,k)\}\right|}{\sqrt{\mathbb{E}\{|X_q(n,k)|^2\}\mathbb{E}\{|X_{q'}(n,k)|^2\}}} \tag{16}$$

where $w_{q,q'}(n,k)$ measures the instantaneous coherence between channels q and q', $\alpha \in [0.5, 1]$ controls the strength of the weighting, and λ is a truncation parameter chosen in proportion to the median magnitude of the cross-power spectrum to suppress unreliable bins.

Compared with the standard PHAT, the proposed $\tilde{R}_{q,q'}(n,k)$ retains partial amplitude information to mitigate symmetry ambiguity, while the truncation term prevents low-energy bins from being artificially amplified. This formulation provides a more robust phase-difference representation for DOA estimation in low-SNR scenarios.

After constructing the inter-channel phase differences, the next step is to match them with the theoretical propagation delays determined by the array geometry in order to estimate the source direction. Consider a uniform grid of candidate directions (θ, φ) in space. For a microphone pair (q, q'), the theoretical time delay can be expressed as

$$\tau_{q,q'}(\theta, \varphi) = \frac{1}{c}(\mathbf{r}_q - \mathbf{r}_{q'})^{\top}\mathbf{u}(\theta, \varphi) \tag{17}$$

where $\mathbf{r}_q$ and $\mathbf{r}_{q'}$ are the position vectors of the q-th and q'-th microphones, $\mathbf{u}(\theta, \varphi)$ denotes the unit direction vector corresponding to azimuth θ and elevation φ, and c is the speed of sound.

Given a candidate direction (θ, φ), the consistency between the measured and theoretical phase differences for the j-th microphone pair can be evaluated by

$$\Psi_j(n,k;\theta,\varphi) = \tilde{R}_{q,q'}(n,k)\,\exp\big(-j2\pi f_k \tau_{q,q'}(\theta,\varphi)\big) \tag{18}$$

where $\tilde{R}_{q,q'}(n,k)$ denotes the improved cross-power spectrum term introduced in (13), and f_k is the frequency of the k-th bin. When the assumed direction (θ, φ) matches the true source, the phase difference becomes consistent and $\Psi_j(n,k;\theta,\varphi)$ approaches its maximum.

To improve robustness under low-SNR conditions, we aggregate the measurements over the time frames $\mathcal{T}$, frequency bins $\mathcal{K}$, and microphone pairs $\mathcal{J}$. The final angular spectrum is defined as

$$\Psi(\theta,\varphi) = \sum_{j\in\mathcal{J}}\sum_{n\in\mathcal{T}}\sum_{k\in\mathcal{K}} \Psi_j(n,k;\theta,\varphi) \tag{19}$$

The DOA is then estimated by searching for the maximum peak of the angular spectrum:

$$(\hat{\theta}, \hat{\varphi}) = \arg\max_{\theta,\varphi} |\Psi(\theta,\varphi)| \tag{20}$$

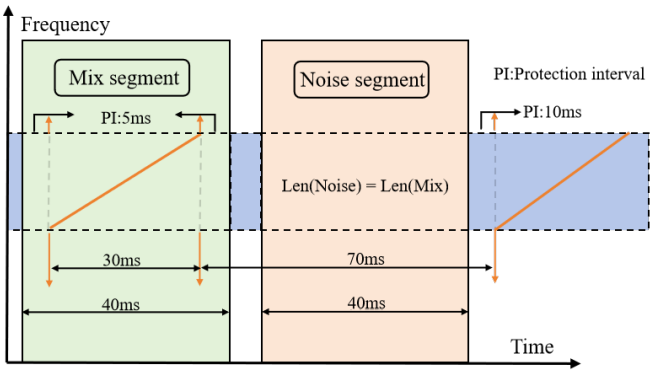


Fig. 5: **FDMA frequency allocation across anchors.** The mix segment (green area) contains the superposition of the transmitted chirp and environmental noise, while the noise segment (orange area) represents the background noise with the same duration as the mix segment, which is used for noise modeling or covariance estimation. A short protection interval (PI = 5 ms) is applied before and after each mix segment, and a longer PI (10 ms) is placed around the noise segment to ensure signal isolation and avoid interference between adjacent chirps.

C. Noise Subtraction with Pre-Chirp Noise Estimation

Before performing noise suppression, it is necessary to detect the coarse time-of-arrival (ToA) of each chirp transmitted by the anchors. Since the chirp waveform is known a priori, we employ a matched filter whose impulse response is the time-reversed conjugate of the reference chirp $s_{\text{ref}}(t)$. For the received signal $x(t)$, the matched filter output is given by

$$y(\tau) = \int x(t)\, s^*_{\text{ref}}(t-\tau)\, dt \tag{21}$$

which is equivalent to the cross-correlation between the received signal and the reference chirp. The coarse ToA of the ℓ-th chirp is then obtained as

$$\hat{\tau}_\ell = \arg\max_{\tau} |y(\tau)| \tag{22}$$

where $\hat{\tau}_\ell$ denotes the coarse arrival time of the ℓ-th chirp.

With $\hat{\tau}_\ell$ determined, we define two non-overlapping STFT windows: (i) a *signal+noise* window starting from $\hat{\tau}_\ell$ with duration T_{sig}, and (ii) a *noise-only* window of length Δ_{pre} located immediately before $\hat{\tau}_\ell$, with a short guard interval g reserved to prevent signal leakage, as shown in Fig. 5.

As defined in (15) and (16), the time-frequency consistency measure $\Psi_j(n,k;\theta,\varphi)$ incorporates the normalized cross-power spectrum and the theoretical delay $\tau_{q,q'}(\theta,\varphi)$.

By aggregating over time, frequency, and microphone pairs, we obtain the angular spectra of the mixture and the drone noise, respectively:

$$\Psi^{\text{mix}}_\ell(\theta,\varphi) = \sum_{j\in\mathcal{J}} \sum_{n\in\mathcal{T}^{\text{sig}}_\ell} \sum_{k\in\mathcal{K}} \Psi_j(n,k;\theta,\varphi) \tag{23}$$

$$\Psi^{\text{noi}}_\ell(\theta,\varphi) = \sum_{j\in\mathcal{J}} \sum_{n\in\mathcal{T}^{\text{noi}}_\ell} \sum_{k\in\mathcal{K}} \Psi_j(n,k;\theta,\varphi) \tag{24}$$

A straightforward subtraction can then be performed to obtain the cleaned angular spectrum:

$$\Psi^{\text{clean}}_\ell(\theta,\varphi) = \beta_\ell \Psi^{\text{mix}}_\ell(\theta,\varphi) - \Psi^{\text{noi}}_\ell(\theta,\varphi) \tag{25}$$

where $\beta_\ell \geq 0$ compensates for energy mismatch between the two windows. In practice, β_ℓ may be estimated via least squares or median ratio methods. To avoid over-subtraction, a non-negativity constraint is enforced:

$$\Psi^{\text{clean}}_\ell(\theta,\varphi) \leftarrow \max\big(\Psi^{\text{clean}}_\ell(\theta,\varphi),\, 0\big) \tag{26}$$

GEVD-Based Reliability Weighting: In the above subtraction process, the scaling factor β_ℓ is introduced to compensate for the energy mismatch between the signal and noise windows. However, directly adopting a fixed or empirically estimated β_ℓ still has certain limitations. On the one hand, the energy difference may vary dynamically across different time frames and frequency bins; on the other hand, when some bins are dominated by drone noise, using a uniform scaling factor may cause over- or under-subtraction, thereby introducing spurious peaks into the angular spectrum. To further address this issue, we incorporate a time-frequency reliability weighting based on *generalized eigenvalue decomposition (GEVD)*, which adaptively enhances the accumulation process of the angular spectrum.

Specifically, the reference noise covariance matrix is estimated for each frequency bin within the noise-only window as

$$\mathbf{R}^{(\ell)}_{\text{noi}}(k) = \frac{1}{|\mathcal{T}^{\text{noi}}_\ell|} \sum_{n\in\mathcal{T}^{\text{noi}}_\ell} \mathbf{X}(n,k)\mathbf{X}^H(n,k) \tag{27}$$

where $\mathbf{x}(n,k) = [X_1(n,k), X_2(n,k), \ldots, X_M(n,k)]^\top$ denotes the array observation vector.

Similarly, in the signal window we compute the instantaneous mixture covariance:

$$\mathbf{R}^{(\ell)}_{\text{mix}}(n,k) = \mathbf{X}(n,k)\mathbf{X}^H(n,k) \tag{28}$$

For numerical stability, the covariance matrices are Hermitianized and diagonally loaded:

$$\mathbf{R} \leftarrow \tfrac{1}{2}(\mathbf{R}+\mathbf{R}^H) + \delta\tfrac{\text{tr}(\mathbf{R})}{M}\mathbf{I}, \quad \delta \sim 10^{-3} \tag{29}$$

At each time-frequency point (n,k), we solve the generalized eigenvalue problem:

$$\mathbf{R}^{(\ell)}_{\text{mix}}(n,k)\mathbf{v} = \gamma_\ell(n,k)\, \mathbf{R}^{(\ell)}_{\text{noi}}(k)\mathbf{v} \tag{30}$$

and retain the largest generalized eigenvalue $\gamma_{\max,\ell}(n,k)$. This value quantifies the power ratio in the most favorable direction, hence distinguishing "signal+noise" bins from "noise-only" bins. When $\gamma_{\max} \approx 1$, the bin is noise-dominated, whereas $\gamma_{\max} \gg 1$ indicates strong signal presence.

The GEVD-based reliability weight is defined as

$$\beta_\ell(n,k) = \left(\frac{\max\{1,\gamma_{\max,\ell}(n,k)\}-1}{\max\{1,\gamma_{\max,\ell}(n,k)\}}\right)^\rho, \quad \rho\in(0.5,1] \tag{31}$$

where ρ controls the compression of the weighting curve. Thus, noise-dominated bins are assigned weights close to zero, while signal-dominated bins are close to one.

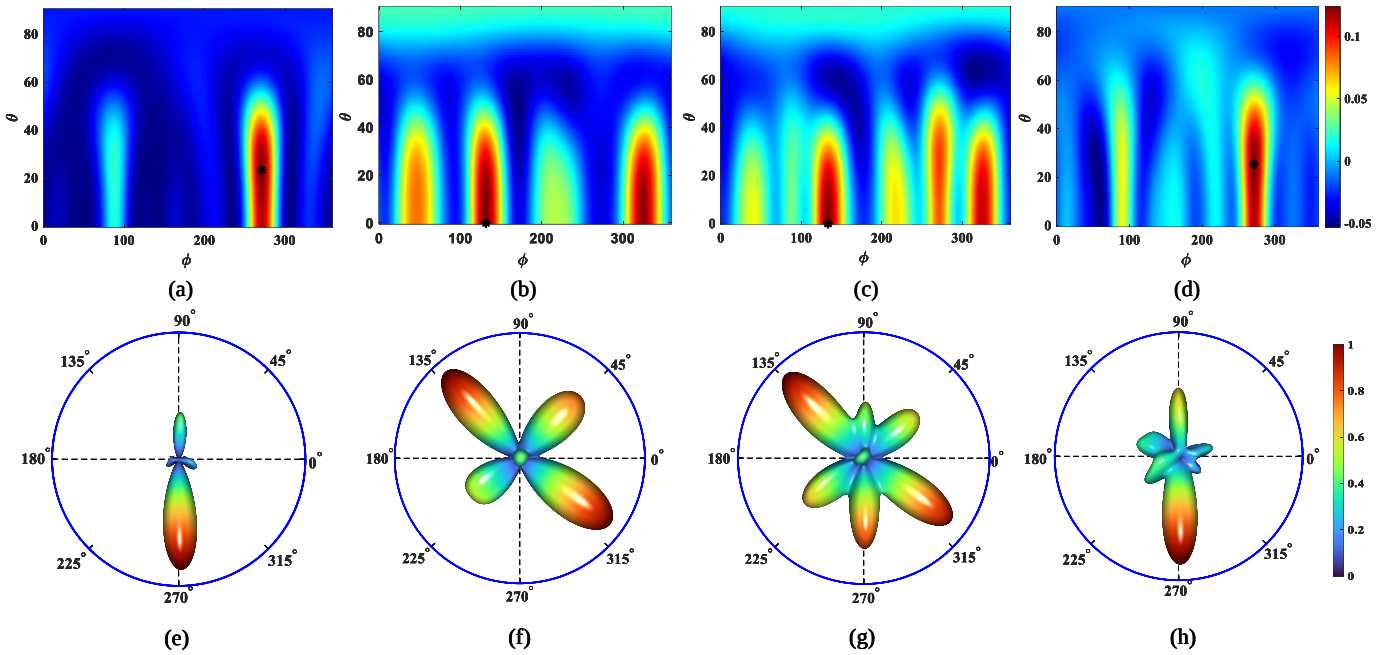


Fig. 6: **Angle estimation results of the proposed RAS system.** (a) Angle spectrum of the pure chirp signal. (b) Angle spectrum of the pure noise signal. (c) Angle spectrum of the mixed signal. (d) Angle spectrum of the denoised signal. (e) DOA radiation pattern of the pure chirp signal. (f) DOA radiation pattern of the pure noise signal. (g) DOA radiation pattern of the mixed signal. (h) DOA radiation pattern of the denoised signal.

The weighted mixture angular spectrum is then given by

$$\Psi_\ell^{\text{mix},\beta_\ell}(\theta,\varphi) = \sum_{j\in\mathcal{J}} \sum_{n\in\mathcal{T}_\ell^{\text{sig}}} \sum_{k\in\mathcal{K}} \beta_\ell(n,k)\, \Psi_j(n,k;\theta,\varphi) \quad (32)$$

The cleaned angular spectrum is updated as

$$\Psi_\ell^{\text{clean}}(\theta,\varphi) = \Psi_\ell^{\text{mix},\beta_\ell}(\theta,\varphi) - \Psi_\ell^{\text{noi}}(\theta,\varphi) \quad (33)$$

with non-negativity enforced:

$$\Psi_\ell^{\text{clean}}(\theta,\varphi) \leftarrow \max\big(\Psi_\ell^{\text{clean}}(\theta,\varphi),\, 0\big) \quad (34)$$

This GEVD-based weighting mechanism, compared with conventional β_ℓ-based subtraction, better captures the variability across time-frequency bins and effectively suppresses spurious peaks and symmetry ambiguities under low-SNR conditions, thereby improving the robustness and resolution of DOA estimation.

As shown in Fig. 6, the proposed RAS algorithm maintains high localization accuracy even under low-SNR conditions. The angle-spectrum and DOA radiation plots indicate that RAS effectively suppresses UAV self-noise and achieves precise chirp-source localization, demonstrating its strong robustness in realistic indoor acoustic environments.

D. UAV Positioning and Heading Estimation

During indoor flight, the UAV may experience attitude variations such as roll and pitch, which can slightly affect the incident direction of the acoustic signals received by the onboard microphone array. However, since the array has a planar geometry, the phase differences across microphones are mainly determined by the projection of the incoming wave direction onto the array plane. As a result, variations in roll and pitch typically induce only minor changes in the projected azimuth direction.

In contrast, the observed azimuth angles are more significantly affected by the UAV yaw rotation during heading changes. Therefore, to recover the UAV position in the world coordinate frame, the UAV yaw angle must be explicitly considered during localization.

Conventional AOA localization approaches usually measure the discrepancy between the theoretical and observed bearings in a scalar form. However, such representations are prone to angular discontinuities (e.g., the equivalence of $-\pi$ and π) and nonlinear accumulation of errors. To improve robustness and maintain geometric consistency, we reformulate the estimation by computing the wrapped angular deviation between the theoretical and observed bearings.

Let the i-th anchor be located at $\mathbf{b}_i = [x_i,\, y_i]^\top$, and denote the UAV state by the planar position $\mathbf{p} = [x,\, y]^\top$ and the heading angle ϕ, which represents the rotation from the UAV body frame to the world frame. The observed AOA in the body frame is θ_i^{obs}. The theoretical bearing angle from $\mathbf{p}$ to $\mathbf{b}_i$ in the world frame is given by

$$\theta_i(\mathbf{p}) = \text{atan2}(y - y_i,\, x - x_i) \quad (35)$$

The residual angular error is defined as the wrapped difference

$$\Delta\theta_i(\mathbf{p},\phi) = \text{wrap}\left(\theta_i(\mathbf{p}) - (\theta_i^{\text{obs}} + \phi)\right) \quad (36)$$

where $\text{wrap}(\cdot)$ maps an angle to $[-\pi,\pi)$ to avoid discontinuity. The robust cost function is then defined using the Huber

loss:

$$\mathcal{L}_{\mathrm{AOA}}(\mathbf{p},\phi)=\sum_{i=1}^{M}\rho_{\kappa}\big(\Delta\theta_i^2(\mathbf{p},\phi)\big) \tag{37}$$

where $\rho_{\kappa}(\cdot)$ denotes the Huber loss function with threshold κ, originally proposed in [50]. The optimal UAV position and heading are obtained by

$$(\hat{\mathbf{p}},\hat{\phi})=\arg\min_{\mathbf{p},\phi}\mathcal{L}_{\mathrm{AOA}}(\mathbf{p},\phi) \tag{38}$$

Due to the nonlinearity of the heading angle ϕ, the optimization problem is prone to local minima. To mitigate this issue, a coarse-to-fine strategy is adopted. A coarse heading search is first conducted over $\phi\in[0,2\pi)$, where for each candidate heading the AOA observations are transformed into the world frame to generate a set of bearing lines originating from the anchor positions. The intersection point of these lines is then estimated via least squares to provide an initial position hypothesis. Among all candidate headings, the one yielding the lowest cost $\mathcal{L}_{\mathrm{AOA}}(\mathbf{p},\phi)$ is selected as the global initialization. Starting from this initial estimate, the parameters are refined by applying the Nelder–Mead simplex method [51] to minimize $\mathcal{L}_{\mathrm{AOA}}(\mathbf{p},\phi)$, which ensures convergence stability without requiring gradient information.

IV. Implementation and Evaluation

The proposed LUNA localization system is built from commodity acoustic components. On the transmitter side, we deploy a set of customized, spatially distributed loudspeakers (see Fig. 7). Each unit consists of a low-cost paper-cone driver and a simple driver board, designed to meet the required sound pressure level while minimizing cost and integration overhead. In our experiments, we place $M=4$ loudspeakers approximately uniformly within the test area to serve as independent beacons.

The receiver is an eight-element UCA with a radius of $1.6\,\mathrm{cm}$. The array is compact and lightweight, which fits the payload and volume constraints of an onboard UAV platform. An eight-element UCA is chosen to provide nearly uniform azimuthal coverage, improved angular resolution, and lower sidelobes compared to four- or six-element arrays, thereby mitigating angular ambiguity and reducing estimation error in DOA inference. All devices are sampled at $48\,\mathrm{kHz}$ to faithfully capture time- and frequency-domain features. During DOA estimation, we perform a full azimuthal search with a 1° step to balance computational cost and angular resolution.

For comparison, a commercial ultra-wideband (UWB) localization system (LinkTrack) is employed as a baseline [52]. The system operates in a time-of-flight (ToF) ranging mode and consists of multiple fixed anchors and a mobile tag mounted on the UAV. The anchors are deployed around the experimental area following the manufacturer-recommended configuration. ToF-based ranging is adopted in this work due to its improved accuracy and robustness compared with TDOA-based schemes under practical synchronization and multipath conditions, making it a representative choice for state-of-the-art commercial UWB localization systems.

Audio and UWB data are recorded within the ROS framework, while motion-capture (MoCap) data are collected by an external system; all timestamps are aligned offline for subsequent analysis.

A. Estimation accuracy of different DOA algorithms

UAVs generate substantial onboard self-noise during flight, which makes it increasingly difficult for the receiver to reliably capture acoustic beacon signals as the SNR decreases. Meanwhile, AOA based localization methods are inherently sensitive to angular estimation errors, such that degradation in SNR directly translates into instability and loss of reliability in the estimated directions.

Importantly, for active acoustic sensing systems deployed on UAV platforms, the effective operating range is not solely determined by geometric distance. Instead, it is jointly influenced by multiple factors, including propagation distance, environmental attenuation, platform-dependent self-noise, and the transmitted signal power. These factors collectively manifest as variations in the received SNR. As a result, SNR provides a more general and physically meaningful indicator of system operating conditions than distance alone.

To this end, we conduct comparative experiments across multiple SNR levels to characterize the performance differences and robustness of representative DOA algorithms in noisy, multipath-prone environments, thereby providing insight into their effective operating regimes.

1) Experimental Design: To evaluate angular estimation performance under different SNR conditions, we design both static and dynamic scenarios. Unlike the common practice of synthetically adding Gaussian noise, we modulate the *actual* received SNR by adjusting the source-UAV distance, which better reflects real-world reception dominated by UAV self-noise. In the static experiments, the UAV hovers at a fixed position with a stable attitude, while the acoustic source operates at a fixed transmission power and is deployed at increasing distances from the UAV, ranging from 5 m to 20 m. The UAV hovers at a height of approximately $1.5\,\mathrm{m}$ above the floor, while the anchor nodes are installed at a fixed height of $2.5\,\mathrm{m}$. Under the current system configuration, the effective SNR measured from the signals recorded by the onboard microphone array ranges approximately from -15 dB to -30 dB. Only a single acoustic source is used in the static experiments. To create different source–receiver geometries, the same source is physically moved to different positions around the UAV rather than activating multiple anchors simultaneously. For each distance configuration, five discrete source positions are selected to account for angular diversity. At each position, multiple frames of signals are recorded to improve statistical reliability (see Fig. 8). In the dynamic experiments, the acoustic source remains fixed at a distance of approximately 15 m from the UAV, while the UAV executes a continuous flight. During the flight, the effective received SNR varies within the range of approximately -30 dB to -25 dB due to motion-induced fluctuations and self-noise. The onboard microphone array continuously records multi-frame signals, which are subsequently processed offline to estimate the AOA. This enables an evaluation of angular accuracy, trajectory continuity, and overall robustness under the combined effects of low SNR and platform motion.

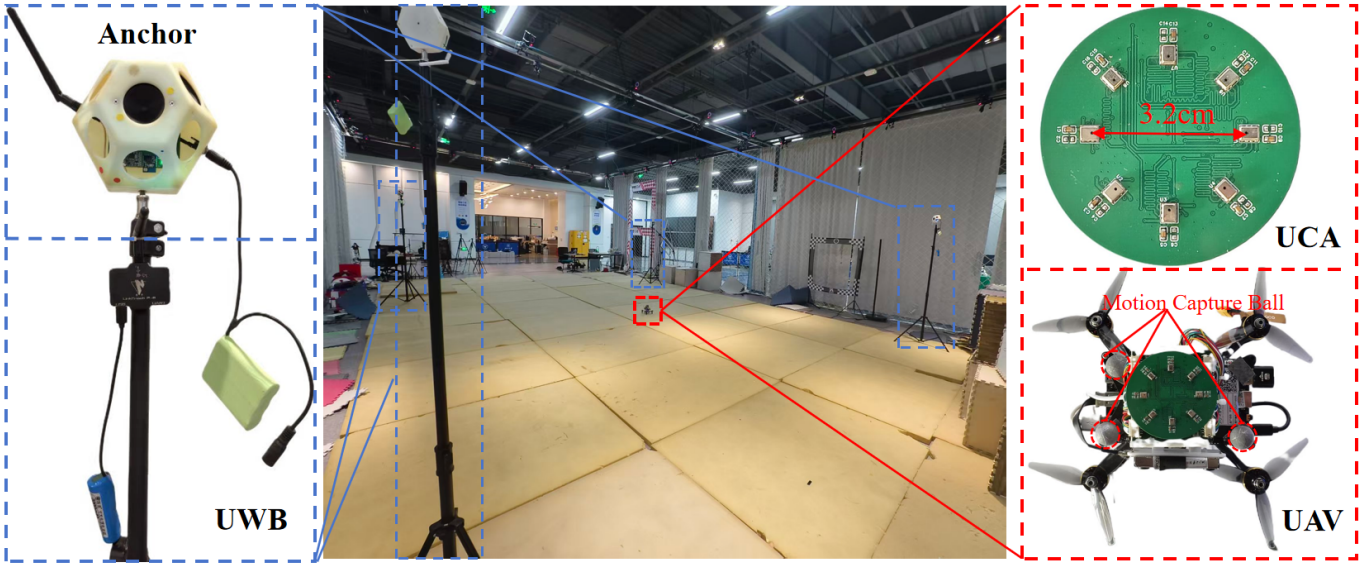


Fig. 7: **Experimental setup of the indoor UAV localization system.** Four acoustic anchors (left) are deployed around the motion capture area to transmit FDMA-LFM signals, while a UWB module provides UWB positioning data. The UAV carries a UCA (diameter 3.2 cm) for signal reception and DOA estimation. Motion capture markers mounted on the UAV are used to obtain the ground-truth position during experiments.

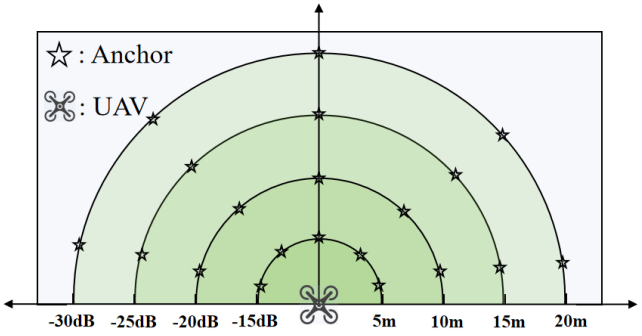


Fig. 8: **Schematic diagram of the static experimental setup for the LUNA system.** The acoustic source is positioned at distances ranging from 5 m to 20 m from the UAV, and the corresponding effective SNR measured from the onboard microphone array is approximately -15 dB to -30 dB.

2) Static Experiment Results: The chirp signal used in the static experiments has a duration of 30 ms and a transmission period of 100 ms. Since each chirp produces one DoA estimate, the effective update rate of the system is approximately 10 Hz. The static experimental results shown in Fig. 9 are obtained under the deployment illustrated in Fig. 8. Specifically, each reported SNR level corresponds to a fixed source–UAV distance, represented by the concentric circles in Fig. 8. The acoustic anchor is installed at a fixed height of approximately 2.5 m above the ground, while the UAV hovers at about 1.5 m during the measurements. For each distance configuration, measurement data are collected at five different angular positions uniformly distributed around the UAV. This design introduces angular diversity, which helps mitigate bias caused by a single source direction and partially captures the spatial variations introduced by multipath propagation.

As shown in Fig. 9, in the static scenario, as the SNR is gradually reduced from -15 dB to -30 dB, all DOA algorithms exhibit performance degradation, but with markedly different rates and thresholds. At -15 dB, all methods operate stably with root-mean-square error (RMSE) generally within $5°$, and the proposed RAS method achieving less than $2°$. When the SNR drops to -20 dB, generalized cross-correlation with phase transform (GCC-PHAT) rapidly deteriorates with RMSE approaching $100°$, while steered response power with phase transform (SRP-PHAT) shows a notable bias of about $30°$; in contrast, both MUSIC and RAS remain around the $10°$ level. Pushing the SNR further down to -25 dB and -30 dB leads GCC-PHAT and SRP-PHAT to effectively fail (RMSE $>$ $200°$, nearing randomness), whereas MUSIC becomes highly volatile with RMSE exceeding $100°$. By comparison, RAS sustains an RMSE below $10°$ even at -30 dB, demonstrating substantial robustness and an extended usability boundary under extremely low-SNR and strong multipath conditions.

Conventional subspace-based methods such as MUSIC rely on a clear distinction between signal and noise subspaces; however, at very low SNR or with limited snapshots, subspace leakage and spectrum broadening occur, leading to unstable estimates. In contrast, the proposed RAS method incorporates robust phase-consistency weighting and a sparsity-inducing prior, which jointly enhance main-lobe concentration and suppress spurious peaks, yielding a distinct spectral maximum even under extreme noise conditions. Correlation-based GCC-PHAT and energy-focusing SRP-PHAT further degrade in such regimes: the former loses its correlation peak as noise dominates, while the latter accumulates noise energy across bins, producing false peaks and angular bias. Overall, RAS maintains high angular resolution and robustness by combining

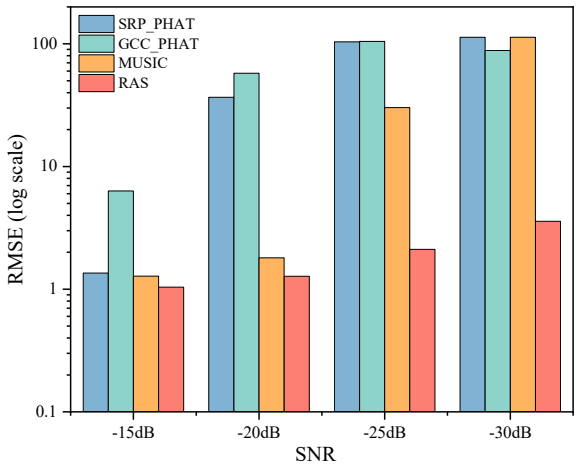


Fig. 9: **RMSE comparison of different DOA estimation algorithms under various SNR conditions.** The main plot shows the overall RMSE variation of SRP-PHAT, GCC-PHAT, MUSIC, and the proposed RAS across SNR levels from $-15\,\mathrm{dB}$ to $-30\,\mathrm{dB}$, where the Y-axis is displayed in logarithmic scale.

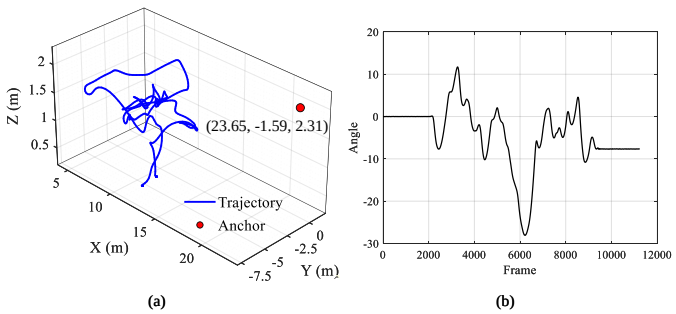


Fig. 10: **Three-Dimensional Motion Trajectory and Yaw Angle Variation of the UAV Relative to the Base Station.** (a) Three-dimensional flight trajectory of the UAV and the spatial coordinates of the base station, where the flight altitude of the UAV is approximately 1.5 m. (b) Yaw angle variation of the UAV relative to the base station.

phase reliability weighting with sparsity-driven denoising.

Beyond accuracy comparison, the above results also enable an examination of the practical operating boundary of AOA-based localization under the current system implementation. In our experiments, the effective SNR is measured directly from the signals recorded by the onboard microphone array of the UAV. When the effective SNR drops below approximately -30 dB, even the proposed RAS method begins to exhibit a rapid increase in angular uncertainty, with estimation errors no longer remaining stable or bounded across consecutive frames. In this regime, the estimated angles become highly sensitive to noise and multipath perturbations, resulting in pronounced frame-to-frame fluctuations and a loss of directional consistency.

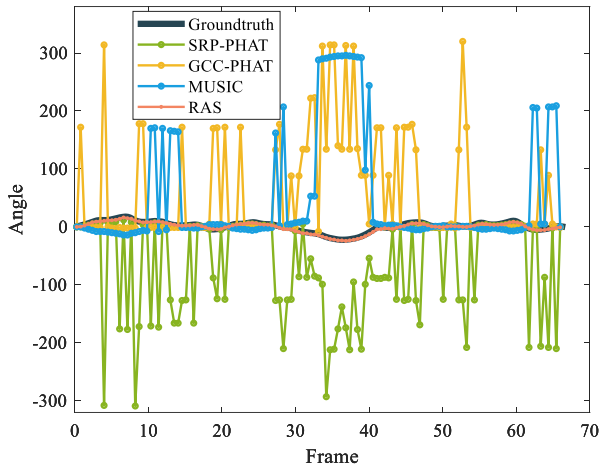
From a localization perspective, such angular degradation has a particularly severe impact on position estimation. In multi-anchor AOA-based triangulation, localization error scales with both the source-to-UAV distance and the angular estimation error; as a result, even moderate angular deviations can lead to substantial spatial misalignment at longer ranges. Therefore, although angular estimates may still be numerically produced when the effective SNR falls below -30 dB, their reliability is insufficient to support meaningful AOA-based localization.In light of this observation, SNR conditions below -30 dB are not further investigated in this work. As shown in Fig. 8, under the current experimental setup, the effective SNR measured from the onboard microphone array decreases to approximately -30 dB when the source-to-UAV distance reaches about 20 m. This distance can thus be regarded as the practical upper bound of the effective operating range of the proposed system.

3) Dynamic Experiment Results: Prior to presenting the dynamic localization results, we first describe the experimental motion conditions to provide sufficient contextual information. As illustrated in Fig. 10(a), the three-dimensional flight trajectory of the UAV and the spatial coordinates of the acoustic anchor are explicitly shown. During the experiment, the UAV followed a free-flight trajectory at an average speed of approximately 1.0 m/s, with a flight altitude around 1.5 m. Fig. 10(b) depicts the yaw angle variation of the UAV relative to the base station over time. This yaw trajectory serves as the ground-truth reference for evaluating the three angular estimation algorithms under dynamic conditions.

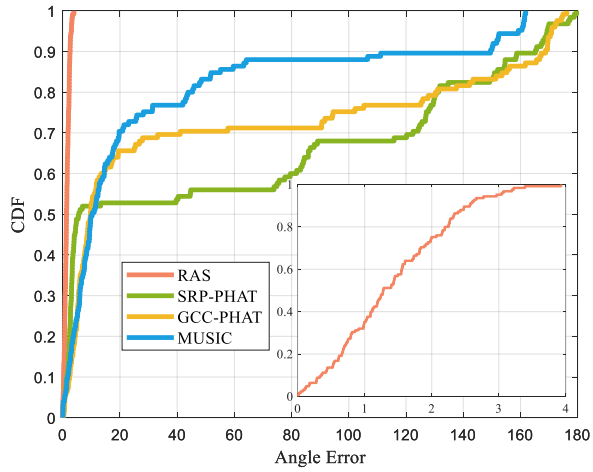
As shown in Fig. 11(a), under dynamic flight with a fixed source and an SNR ranging from $-25\,\mathrm{dB}$ to $-30\,\mathrm{dB}$, the time-evolving angle estimates reveal pronounced differences across methods. GCC-PHAT and MUSIC exhibit frequent offsets and abrupt jumps, with instantaneous errors exceeding $100°$ over multiple segments; SRP-PHAT likewise shows strong time-varying fluctuations. In contrast, the proposed RAS produces a smoother and more continuous angular trajectory that closely follows the ground truth, indicating stronger temporal consistency and disturbance rejection. The cumulative distributions in Fig. 11(b) further quantify these trends: in this low-SNR dynamic regime, the median error (P50) of GCC-PHAT and MUSIC remains above $100°$, suggesting limited practical utility, while SRP-PHAT exhibits a broadly spread error distribution that challenges real-time requirements. By comparison, RAS maintains a 90th-percentile error (P90) below $3°$, with a markedly higher cumulative probability in the small-error region, thereby demonstrating superior stability and robustness under concurrent motion, noise, and multipath.

B. Overall Performance under Indoor Flight Conditions

This section evaluates UAV localization performance under standard indoor conditions. Given the eight-element UCA and its limitations arising from aperture and near-field effects, reliable elevation (pitch) estimation cannot be guaranteed; we therefore restrict our analysis to planar position and yaw attitude. Motion-capture (MoCap) measurements of 2D position and yaw serve as the ground-truth reference. For this performance evaluation, the flight area is set to $6\,\mathrm{m} \times 10\,\mathrm{m}$. We compare the proposed LUNA with a commercial UWB system (LinkTrack) and an acoustic TDOA-based method under both line-of-sight (LoS) and near-wall non-line-of-sight



(a)



(b)

Fig. 11: **DOA estimation results of different algorithms under dynamic UAV flight conditions.** The experimental workspace corresponds to an SNR range of -25 dB to -30 dB. (a) Comparison between the estimated DOA trajectories of SRP-PHAT, GCC-PHAT, MUSIC, and the proposed RAS with the ground-truth motion-capture reference. (b) CDF of the angular error for each algorithm, where the inset provides a zoomed view of the low-error region.

(NLoS) indoor conditions, where the effective signal-to-noise ratio of the received acoustic signal measured by the onboard microphone array is approximately -22 dB.

As shown in Fig. 12(a), we present the trajectory reconstruction results of the three localization methods under LoS indoor conditions, compared against the MoCap ground truth. The proposed LUNA reconstructs the UAV trajectory accurately over the entire flight path; only minor deviations appear in high-curvature segments such as turns and takeoff/landing, and the estimated track remains continuous and smooth. The UWB-based solution achieves a generally high localization accuracy and successfully recovers the global trajectory shape. Most straight-line segments closely align with the ground truth. However, a systematic positional bias can be observed in the latter part of the trajectory, where the estimated path exhibits a consistent offset along the y-axis. The TDOA approach recovers the gross trajectory shape but shows conspicuous jitter and outliers near boundaries and in cluttered areas. This behavior is attributable to strong multipath from wall reflections and scattering: leading or trailing echoes bias the TOA/TDOA estimates and the resulting small timing errors are amplified by the geometric inversion, yielding unstable position solutions.

Fig. 12(b) shows the cumulative distribution functions (CDF) of localization error for the three methods. Overall, the curves exhibit distinct distribution trends, reflecting different accuracy and stability characteristics. The proposed LUNA method produces the leftmost and steepest CDF curve, indicating a high proportion of small localization errors. The localization error at the 50% level is approximately 0.07 m, and the error at the 90% level is 0.16 m. This indicates not only low overall error but also high concentration and consistency in positioning performance. The UWB method presents a right-shifted CDF curve compared with LUNA, while maintaining a comparable overall trend. The localization error at the 50% level is 0.12 m, and the error at the 90% level reaches 0.22 m, indicating a moderately broader error distribution relative to LUNA. In contrast, the TDOA method exhibits the rightmost CDF curve with a pronounced long tail. The localization errors at the 50% and 90% levels increase to 0.40 m and 1.26 m, respectively, highlighting its sensitivity to multipath interference and degraded robustness under low-SNR conditions. The inset of Fig. 9(b) further presents the CDF of the angular error for the LUNA method, showing that 90% of the estimated angles fall within 1° of deviation, which demonstrates its precise directional estimation capability.

As shown in Fig. 13(b), NLoS conditions are constructed by placing obstacles in front of anchors 1 and 3 within the same experimental area as the LoS case. Fig. 13(a) presents the reconstructed flight trajectories of the three localization methods under this NLoS scenario, where the light shaded region highlights the area with dominant NLoS propagation. From the trajectory results, the proposed LUNA system exhibits a noticeable global drift in the NLoS region. This behavior is mainly attributed to increased angular estimation errors at anchors 1 and 3, where partial occlusion and strong multipath reflections degrade the reliability of DoA measurements. The TDOA-based approach suffers from a more severe degradation, with a large number of missing position estimates in the NLoS region, indicating its high sensitivity to corrupted time-of-arrival information under multipath interference. In comparison, the UWB system shows only a mild positional bias in the NLoS area. This can be partially explained by the fact that UWB radio signals exhibit a certain degree of penetration capability and can propagate through low-density obstacles such as cardboard panels, thereby preserving a subset of valid ranging measurements.

To further quantify the impact of NLoS propagation, the CDFs of the localization errors under NLoS conditions are presented in Fig. 14. Compared with the corresponding LoS results, it can be observed that UWB experiences only a moderate degradation, with the error distribution remaining

relatively concentrated in the low-error region. In contrast, both LUNA and the TDOA-based method exhibit a noticeable rightward shift in the CDF curves under NLoS conditions, indicating a significant increase in localization errors. The performance deterioration of LUNA is primarily associated with degraded angular estimation reliability at partially occluded anchors, while the TDOA approach shows the most severe degradation due to its strong sensitivity to multipath-induced time-delay corruption. These statistical observations are consistent with the trajectory-level analysis shown in Fig. 13(a).

The quantitative results summarized in Tables II and III further corroborate these observations. Under NLoS conditions, both the positioning error and angular error of the LUNA system increase compared with the LoS case. Specifically, the mean positioning error rises from 0.089 m to 0.283 m, while the mean angular error increases from $0.56°$ to $2.94°$. Despite this performance degradation, LUNA consistently outperforms the TDOA-based method. In comparison, the UWB trajectory exhibits relatively small differences between LoS and NLoS conditions. However, under LoS conditions, the localization accuracy of the UWB system remains lower than that achieved by LUNA. Notably, the increase in LUNA's error remains bounded, indicating that the system does not suffer from catastrophic failures in NLoS environments. This robustness stems from the multi-anchor fusion strategy employed in LUNA. Under NLoS conditions, the system evaluates the temporal fluctuations of DoA estimates across all anchors and selects a subset of at least three anchors with smaller angular variations for final position estimation. By adaptively suppressing unreliable measurements, LUNA mitigates the impact of localized NLoS effects and maintains stable localization performance even when a subset of anchors is severely degraded.

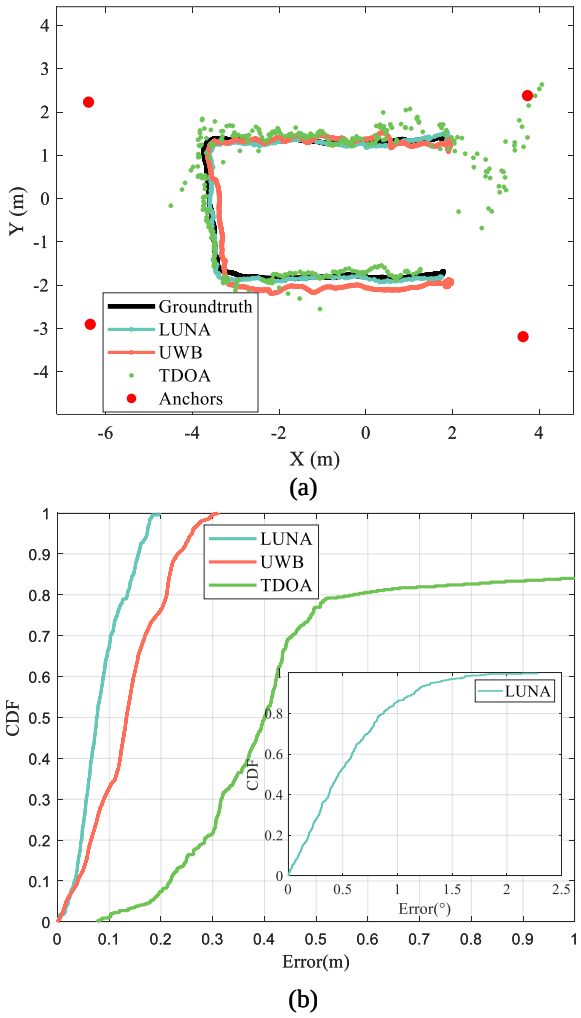


Fig. 12: **Localization results of different positioning algorithms.** (a) Trajectory comparison of different localization methods. (b) CDF of the position error for all methods, with the inset showing the angular error CDF of the LUNA algorithm.

C. Impact of SNR

To evaluate the robustness of the proposed system under different SNR conditions, three experimental scenarios with increasing spatial scales were designed. Specifically, the flight areas were set to $6\,\mathrm{m} \times 6\,\mathrm{m}$, $6\,\mathrm{m} \times 10\,\mathrm{m}$, and $6\,\mathrm{m} \times 18\,\mathrm{m}$, respectively. For all scenarios, the transmitting signal parameters, including the chirp bandwidth and duration, were kept identical. By analyzing the signals received by the onboard microphone array, the overall effective SNR of the valid acoustic signal was estimated to be approximately -16 dB, -22 dB, and -28 dB for the three scenarios, respectively.

As shown in Fig. 15 and Table IV, for the proposed LUNA system, the localization error gradually increases as the SNR decreases. Specifically, the mean localization error increases from 0.077 m at approximately -16 dB to 0.104 m at approximately -28 dB, while remaining at the centimeter level overall. This indicates that LUNA maintains good stability under low-SNR conditions.

In contrast, the localization error of the UWB method remains nearly constant across the three experimental area configurations, with its RMSE staying around 0.14 m, suggesting that its performance is largely independent of acoustic SNR variations. However, in all tested scenarios, the overall localization accuracy of UWB is consistently lower than that of LUNA, with the performance gap becoming more pronounced in moderate and low-SNR indoor environments. The TDOA-based method, on the other hand, is highly sensitive to acoustic noise, and its localization error increases rapidly as the SNR decreases, eventually degrading to meter-level errors under the lowest SNR condition, indicating poor robustness.

It is worth noting that, due to the lower propagation loss of radio-frequency signals, UWB systems generally support a longer effective localization range than acoustic-based approaches. However, this advantage typically comes at the expense of higher hardware cost. By contrast, the proposed LUNA system is constructed from low-cost, commodity acoustic components and is able to achieve superior localization accuracy compared to UWB within typical indoor operating ranges. Overall, these results demonstrate that LUNA achieves a favorable balance among cost, accuracy, and robustness for indoor UAV localization, while maintaining reliable localiza-

TABLE II: Positioning Performance of Different Localization Algorithms Under LoS and NLoS Conditions.

Scenario	UWB			TDOA			LUNA		
	Mean (m)	Median (m)	90% (m)	Mean (m)	Median (m)	90% (m)	Mean (m)	Median (m)	90% (m)
LoS	0.128	0.130	0.218	0.643	0.495	1.264	**0.089**	**0.061**	**0.186**
NLoS	0.110	0.158	0.260	1.137	1.138	1.883	0.283	0.291	0.512

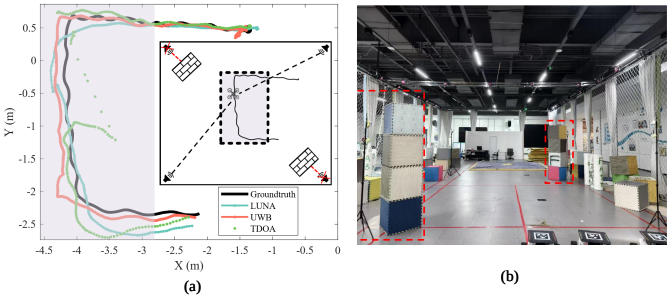


Fig. 13: **Localization performance under NLoS conditions.** (a) Trajectory comparison of different localization methods. (b) NLoS experimental setup.

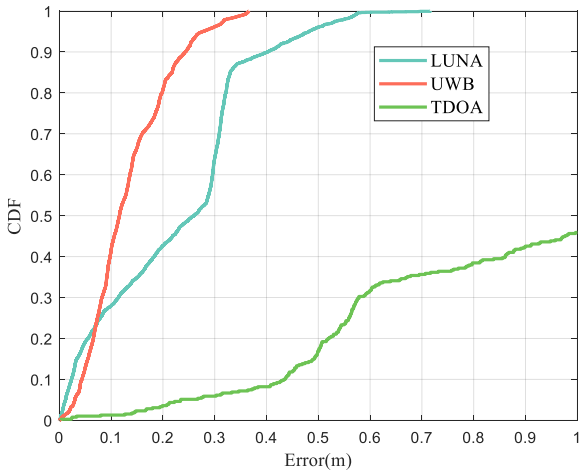


Fig. 14: **Localization results of different algorithms under NLoS conditions.**

tion performance under low-SNR conditions.

Table V summarizes the angular estimation results of the LUNA system under different SNR conditions. The mean angular error remains below $0.65°$, and the RMSE varies only slightly from $0.62°$ to $0.82°$ as the SNR decreases from $-16\,\mathrm{dB}$ to $-28\,\mathrm{dB}$. The median and 90th-percentile values also remain nearly unchanged, confirming that the proposed system achieves sub-degree directional accuracy across all noise levels.

The stability of LUNA arises from its phase-consistency–weighted cross-power spectrum and time–frequency reliability weighting, which together emphasize coherent signal bins and suppress unreliable frequency components. It should also be noted that the experiments were not extended to lower SNR levels, since

TABLE III: Angular Estimation Performance of the LUNA System Under LoS and NLoS Conditions.

Number	Mean (°)	RMSE (°)	Median (°)	90% (°)
LoS	0.56	0.71	0.42	1.21
NLoS	2.94	3.46	2.81	4.18

TABLE IV: Positioning Performance of Different Localization Algorithms Under Varying SNR Conditions.

SNR	Method	Mean (m)	Median (m)	90% (m)
-16 dB ($6\,\mathrm{m} \times 6\,\mathrm{m}$)	TDOA	0.491	0.456	0.870
	UWB	0.119	0.117	0.211
	LUNA	**0.077**	**0.064**	**0.164**
-22 dB ($6\,\mathrm{m} \times 10\,\mathrm{m}$)	TDOA	0.643	0.495	1.264
	UWB	0.128	0.130	0.218
	LUNA	**0.089**	**0.061**	**0.186**
-28 dB ($6\,\mathrm{m} \times 18\,\mathrm{m}$)	TDOA	0.738	0.661	1.384
	UWB	0.126	0.133	0.233
	LUNA	**0.104**	**0.090**	**0.217**

at such extreme noise conditions, the coarse ToA estimation of the chirp signal becomes highly unreliable. The resulting large timing deviations make it difficult to accurately segment the signal window, thereby preventing consistent localization performance.

D. Impact of Number of Anchors

In practical deployment scenarios, the localization network may be affected by environmental occlusions, hardware malfunctions, or communication interruptions, which can cause certain base stations to become unavailable or significantly degrade in signal quality. Under such circumstances, the number of effective base stations participating in localization decreases, potentially impacting the overall positioning accuracy and system stability. To evaluate the robustness and resilience of the proposed system under these conditions, we consider a $6\,\mathrm{m} \times 10\,\mathrm{m}$ indoor scenario in which one base station is unavailable, emulating real-world situations where a station is blocked or fails. The localization performance under this reduced-station configuration was then analyzed and compared with the full-station setup to verify the algorithm's stability and error tolerance when subjected to partial base station failures.

As shown in Fig. 16, the RMSE comparison of the three localization algorithms under normal and reduced anchor conditions is illustrated. It can be observed that when one effective anchor is removed, the RMSE of the proposed LUNA algorithm slightly increases from $0.119\,\mathrm{m}$ to $0.172\,\mathrm{m}$,

TABLE V: Angular Estimation Performance of the LUNA System Under Varying SNR Conditions.

SNR	Mean (°)	RMSE (°)	Median (°)	90% (°)
-16dB	0.51	0.62	0.44	1.10
-22dB	0.56	0.71	0.42	1.21
-28dB	0.62	0.82	0.51	1.32

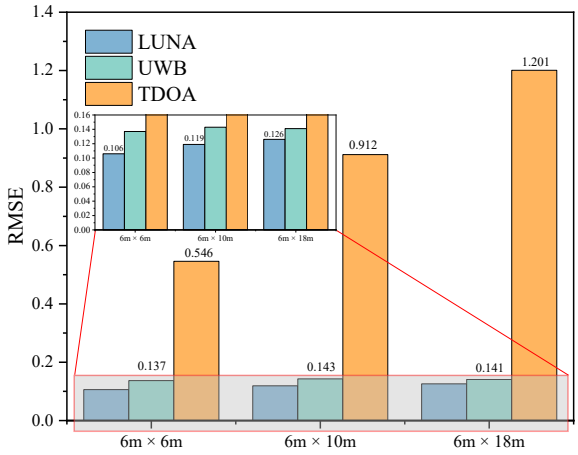


Fig. 15: **RMSE comparison of different localization algorithms under varying SNR conditions.**

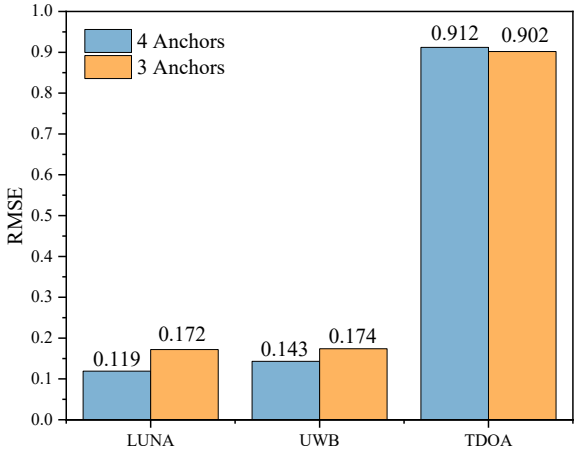


Fig. 16: **RMSE comparison of different localization algorithms under different numbers of anchors.**

TABLE VI: Angular Estimation Performance of the LUNA System Under Different Numbers of Anchors.

Number	Mean (°)	RMSE (°)	Median (°)	90% (°)
4 Anchors	0.56	0.71	0.42	1.21
3 Anchors	1.04	1.36	0.81	2.18

while still maintaining high precision. This indicates that the proposed algorithm exhibits strong robustness against anchor loss. In contrast, the UWB-based localization method shows a slight increase in RMSE from 0.143 m to 0.174 m under anchor reduction, suggesting a relatively stable localization performance.

It is worth noting that the TDOA-based method exhibits a slight decrease in RMSE, from 0.912 m to 0.902 m, after one anchor is removed. This is because the excluded anchor in the original four-anchor configuration introduced larger time-of-arrival measurement errors, as its signal delay characteristics were more affected by multipath and reflections. Consequently, removing this anchor improved the overall consistency of time-delay estimation, thereby statistically reducing the overall root-mean-square error.

As shown in Table VI, the statistical results of azimuth estimation errors of the LUNA system under different anchor configurations are presented. When four anchors are used, the mean error is 0.56° and the RMSE is 0.71°, achieving sub-degree accuracy sufficient for high-precision joint attitude and localization estimation. When one anchor is removed, the mean error increases to 1.04° and the RMSE rises to 1.36°, while the median and 90% confidence errors increase to 0.81° and 2.18°, respectively.

Although the overall angular estimation error increases slightly with fewer anchors, the system still maintains an accuracy better than 2°. This demonstrates that the proposed LUNA algorithm can effectively suppress phase noise accumulation and geometric degradation, even when the array geometry constraints are weakened. These results further validate the robustness and anti-degradation capability of the system under complex deployment conditions, ensuring reliable localization performance in practical scenarios.

E. Time Complexity Analysis

In the system implementation, targeted optimizations are introduced to satisfy the real-time requirements of online UAV operation. By analyzing real flight data, it can be observed that under normal flight conditions, the temporal variation of DOA estimates between consecutive frames is typically limited. Therefore, instead of performing a global angular search at each frame, a temporally constrained local angular search strategy based on prior estimates is adopted.

Specifically, at the k-th frame, a local angular search region is constructed by centering on the DOA estimate from the previous frame. The constrained search domain is defined as

$$\mathcal{S}_k = \left\{ (\theta, \phi) \;\middle|\; |\theta - \hat{\theta}_{k-1}| \le \Delta_\theta,\ |\phi - \hat{\phi}_{k-1}| \le \Delta_\phi \right\} \quad (39)$$

where $(\hat{\theta}_{k-1}, \hat{\phi}_{k-1})$ represent the estimated azimuth and elevation angles at frame $k-1$, and Δ_θ and Δ_ϕ denote the predefined angular window sizes along the azimuth and elevation dimensions, respectively.

Moreover, the GEVD computation is restricted to signal-dominated time–frequency bins rather than being uniformly applied over the entire time–frequency plane, which effectively reduces the number of matrix decompositions required per frame. In the AOA constraint solving stage, the temporal continuity of UAV motion is similarly exploited by using the estimated position and heading from the previous frame as the initialization for the current nonlinear optimization. This warm-start strategy allows the optimization to converge directly in the vicinity of a local optimum in most cases, thereby

TABLE VII: Per-frame Runtime of Major Processing Modules

Module	Runtime (s)
Signal detection	0.0098
DOA estimation	0.0623
AOA constraint solving	0.0048
Total	**0.0769**

avoiding repeated global searches and further improving the overall computational efficiency.

To validate the effectiveness of the above optimization strategies in a practical system and to quantify the contribution of each module to the total computational cost, we further perform a detailed runtime analysis of all functional components. All runtime measurements were conducted on a LattePanda 3 Delta onboard computer equipped with an Intel® Celeron® N5105 quad-core processor (up to 2.9 GHz). The per-frame execution times of individual modules are summarized in Table VII. Among all components in the LUNA framework, the angular estimation module constitutes the dominant computational cost, with an average processing time of 62.3 ms per frame under steady-state conditions. This is followed by the signal detection module and the AOA constraint solver, which require approximately 9.8 ms and 4.8 ms per frame, respectively.

It should be noted that the maximum operating update rate of the proposed system is primarily determined by the design of the transmitted acoustic signal, rather than by the computational capability of the algorithm itself. Specifically, the chirp-based acoustic signal adopted in this work follows a fixed transmission cycle with a total period of 0.1 s, corresponding to a system update rate of approximately 10 Hz. As indicated by the measured runtime of individual processing modules, once the system reaches steady-state operation, the total per-frame computational latency is approximately 75–80 ms. This latency comfortably fits within a single signal period, thereby enabling real-time operation at the designed update rate of 10 Hz.

It is further acknowledged that this update rate is lower than that of some UWB-based localization systems, which can typically operate at several tens of hertz or higher. However, for online localization and state estimation of unmanned aerial vehicles, both position and attitude generally evolve in a temporally smooth manner, and stable system operation does not rely on extremely high update frequencies. Under this consideration, a 10 Hz update rate is sufficient to support real-time onboard operation while providing reliable and robust localization performance in practical flight scenarios. In future work, alternative acoustic signal designs will be explored to further increase the achievable update rate while maintaining robustness under low SNR conditions.

V. Conclusion

In this paper, we proposed an active acoustic localization framework, termed **LUNA**. At the front end, the system constructs a robust cross-spectral representation by integrating coherence weighting and PHAT-λ truncation, while the angle-spectrum subtraction and GEVD-based confidence weighting effectively suppress rotor self-noise and multipath interference. At the back end, a joint position–orientation estimation scheme is achieved solely from DoA information by defining wrapped angular residuals to handle the circular nature of angles, introducing a robust penalty function, and applying the Nelder–Mead optimization algorithm. Comprehensive indoor experiments demonstrate that the proposed LUNA framework maintains centimeter-level position accuracy and sub-degree orientation accuracy even under low-SNR conditions, confirming its robustness and practical feasibility in complex acoustic environments.

Compared with previous methods, LUNA introduces a novel indoor localization framework. Although promising results have been achieved, there remains room for further improvement. In future work, we will investigate array geometries with vertical aperture to enhance the joint estimation accuracy of azimuth and elevation, ultimately enabling high-precision 3D localization. In addition, we will further improve the system refresh rate by optimizing the signal design and streamlining the processing pipeline, so as to better support real-time tracking in dynamic scenarios.

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